Power BI basic

1 : What is Power BI and why is it used in businesses?

- Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps businesses collect, analyze, transform, and visualize data using interactive reports and dashboards.

Why is Power BI used in businesses?

1. Data Visualization: Converts raw data into charts, graphs, and dashboards.
2. Better Decision-Making: Helps managers make informed decisions based on data.
3. Data Integration: Combines data from different sources, such as Excel, SQL databases, and cloud services.
4. Real-Time Monitoring: Helps businesses monitor key performance indicators (KPIs) and track business performance.
5. Time-Saving: Automates report generation and reduces manual work.
6. Easy Sharing: Allows users to share reports and dashboards with team members.

Example

A retail company uses Power BI to analyze its sales data. It creates a dashboard showing:

- Total sales by region.
- Best-selling products.
- Monthly sales trends.
- Profit earned by each branch.

Using this dashboard, the manager can identify which regions perform best and which products need more marketing.

2 : Name and explain the three main components of Power BI.

- The three main components of Power BI are:

1. Power BI Desktop

- It is a desktop application used to connect to data sources, clean and transform data, create data models, and build interactive reports.
- It is mainly used by data analysts to prepare and visualize data.
- Example: Creating a sales report in Power BI Desktop using Excel data.

2. Power BI Service

- It is an online, cloud-based platform used to publish, share, and collaborate on reports and dashboards.
- Users can access reports through a web browser and monitor business performance.
- Example: A sales manager views the company's sales dashboard online.

3. Power BI Mobile

- It is a mobile application that allows users to access reports and dashboards on smartphones and tablets.
- It helps users monitor business performance while away from their computers.
- Example: A manager checks daily sales figures on a mobile phone.

3 : Explain the Power BI workflow.

- The Power BI workflow is the step-by-step process of collecting data, transforming it, creating reports, and sharing insights to support business decision-making.
- **Steps in the Power BI Workflow**
- 1. Data Collection: Connect to data sources such as Excel files, SQL databases, CSV files, or cloud services.
- 2. Data Cleaning and Transformation: Use Power Query to remove duplicates, handle missing values, and correct data formats.
- 3. Data Modeling: Create relationships between tables and define calculations using DAX (Data Analysis Expressions).
- 4. Data Visualization: Create charts, graphs, tables, and interactive reports to understand data.
- 5. Publishing Reports: Publish reports from Power BI Desktop to the Power BI Service.
- 6. Sharing and Monitoring: Share dashboards and reports with team members and monitor business performance.
- **Example**
- A company wants to analyze its sales performance. It imports sales data from Excel, cleans the data, creates relationships between sales and product tables, builds charts showing monthly sales, and publishes the report so managers can access it online.

4 : List any four data cleaning tasks that can be performed in Power Query

- Power Query is a data transformation tool in Power BI used to clean and prepare data before analysis.
- The four data cleaning tasks are:
- 1. Removing Duplicates: Removes repeated records from a dataset to avoid incorrect analysis.
- 2. Handling Missing Values: Removes rows with missing values or replaces null values with suitable values.
- 3. Changing Data Types: Converts data into the correct format, such as text, number, date, or currency.

- 4. Removing Errors: Identifies and removes or replaces errors in the data, such as invalid values or conversion errors.
- Example: In a sales dataset, Power Query can remove duplicate sales records, replace missing product names, convert dates into the correct format, and remove rows containing errors.

6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

- Power BI Desktop has three main views that help users manage, visualize, and organize data.

1. Data View

- Definition: Data View displays the data stored in tables within the Power BI data model.
- Purpose: It is used to inspect rows and columns, check data values, and create calculated columns using DAX.
- Example: Viewing a sales table to check product names, prices, and quantities.

2. Report View

- Definition: Report View is the workspace where users create interactive reports and visualizations.
- Purpose: It is used to build charts, graphs, tables, and dashboards to analyze data and present insights.
- Example: Creating a bar chart showing total sales by region.

3. Model View

- Definition: Model View displays the tables in a data model and the relationships between them.
- Purpose: It is used to create and manage relationships between tables and organize the data model.
- Example: Connecting a Sales table with a Products table using Product ID.

Summary

View	Main Purpose
Data View	Inspect and work with table data

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Report View	Create visualizations and reports
Model View	Manage relationships between tables

7 : Discuss the different data sources that Power BI supports.

- Power BI supports various data sources that allow users to collect, analyze, and visualize data from different platforms.
- The main data sources supported by Power BI are:
- 1. Excel Files: Power BI can import data from Excel workbooks for analysis and reporting.
- Example: Analyzing monthly sales data stored in an Excel file.
- 2. Text and CSV Files: Power BI can import data from CSV (Comma-Separated Values) and text files.
- Example: Analyzing customer details stored in a CSV file.
- 3. Databases: Power BI connects to databases such as SQL Server, MySQL, Oracle, and PostgreSQL.
- Example: Retrieving employee and salary records from a SQL database.
- 4. Online Services: Power BI can connect to online platforms and cloud services, such as SharePoint, Salesforce, and Microsoft Azure services.
- Example: Analyzing customer relationship management (CRM) data from Salesforce.
- 5. Web Sources and APIs: Power BI can retrieve data from websites and web APIs.
- Example: Collecting product prices or public statistical data from an online source.
- 6. Cloud Storage: Power BI supports cloud-based data sources such as Azure data services and supported files stored in OneDrive or SharePoint.
- Example: Analyzing business reports stored in OneDrive.